

Sheth L.U.J. & Sir M.V. College Of Science

Subject :- Cyber & Information Security

Practical no 3

Aim :- Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

Code :- CLI

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Practical no 3.py - C:/Users/Vinod Mali/Desktop/Cyber Security/Practical no 3.py (3.14.6)
File Edit Format Run Options Window Help

import hmac
import hashlib

msg = input("Enter the message: ")
key = input("Enter the secret key: ")

# Generate MAC
mac = hmac.new(key.encode(), msg.encode(), hashlib.sha256).hexdigest()

print("\nMessage:", msg)
print("Generated MAC:", mac)

# Verify MAC
verify = input("\nEnter the message to verify: ")

if hmac.compare_digest(mac, hmac.new(key.encode(), verify.encode(), hashlib.sha256).hexdigest()):
    print("MAC Verification Successful: Message is authentic.")
else:
    print("MAC Verification Failed: Message has been altered.")
```

output :-

```
IDLE Shell 3.14.6
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:/Users/Vinod Mali/Desktop/Cyber Security/Practical no 3.py =====
Enter the message: Hello World!
Enter the secret key: 5

Message: Hello World!
Generated MAC: 5cbd54f4b5255af5c159427b88c540f084faa5f2c8f5069dad79ec083c251508

Enter the message to verify: Hello World!
MAC Verification Successful: Message is authentic.

>>>
```

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Code :- GUI

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##GUI
import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

# Generate MAC
def generate_mac():
    msg = message_entry.get("1.0", tk.END).strip()
    key = key_entry.get()

    if not msg or not key:
        messagebox.showerror("Error", "Enter both Message and Secret Key!")
        return

    mac = hmac.new(key.encode(), msg.encode(), hashlib.sha256).hexdigest()

    output.config(state="normal")
    output.delete("1.0", tk.END)
    output.insert(tk.END, f"Message:\n{msg}\n\n")
    output.insert(tk.END, f"Generated MAC:\n{mac}")
    output.config(state="disabled")

# Verify MAC
def verify_mac():
    msg = message_entry.get("1.0", tk.END).strip()
    key = key_entry.get()
    verify = verify_entry.get("1.0", tk.END).strip()

    if not msg or not key or not verify:
        messagebox.showerror("Error", "Fill all fields!")
        return

    mac = hmac.new(key.encode(), msg.encode(), hashlib.sha256).hexdigest()
    verify_mac = hmac.new(key.encode(), verify.encode(), hashlib.sha256).hexdigest()

    if hmac.compare_digest(mac, verify_mac):
        result.config(text="✓ MAC Verification Successful!\nMessage is Authentic.",
            fg="#00FF00")
    else:
        result.config(text="✗ MAC Verification Failed!\nMessage has been Altered.",
            fg="red")

# Main Window
root = tk.Tk()
root.title("Hacker MAC Generator & Verifier")
root.geometry("700x650")
root.configure(bg="black")

title = tk.Label(root,
    text="MESSAGE AUTHENTICATION CODE (MAC)",
    bg="black",
    fg="#00FF00",
    font=("Consolas", 18, "bold"))
title.pack(pady=10)

# Message
tk.Label(root, text="Enter Message:",
    bg="black", fg="#00FF00",
```

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```
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font=("Consolas", 12)).pack()

message_entry = tk.Text(root, height=5, width=70,
                        bg="#101010", fg="#00FF00",
                        insertbackground="#00FF00",
                        font=("Consolas", 11))
message_entry.pack(pady=5)

# Secret Key
tk.Label(root, text="Enter Secret Key:",
        bg="black", fg="#00FF00",
        font=("Consolas", 12)).pack()

key_entry = tk.Entry(root, width=50,
                    bg="#101010", fg="#00FF00",
                    insertbackground="#00FF00",
                    font=("Consolas", 11))
key_entry.pack(pady=5)

# Buttons
btn_frame = tk.Frame(root, bg="black")
btn_frame.pack(pady=10)

tk.Button(btn_frame, text="Generate MAC",
        command=generate_mac,
        bg="#00AA00", fg="black",
        font=("Consolas", 11, "bold"),
        width=18).grid(row=0, column=0, padx=10)

tk.Button(btn_frame, text="Verify MAC",

tk.Button(btn_frame, text="Verify MAC",
        command=verify_mac,
        bg="#00AA00", fg="black",
        font=("Consolas", 11, "bold"),
        width=18).grid(row=0, column=1, padx=10)

# Output
tk.Label(root, text="Generated MAC:",
        bg="black", fg="#00FF00",
        font=("Consolas", 12)).pack()

output = tk.Text(root, height=8, width=70,
                bg="#101010", fg="#00FF00",
                font=("Consolas", 10))
output.pack(pady=5)
output.config(state="disabled")

# Verify Message
tk.Label(root, text="Enter Message to Verify:",
        bg="black", fg="#00FF00",
        font=("Consolas", 12)).pack()

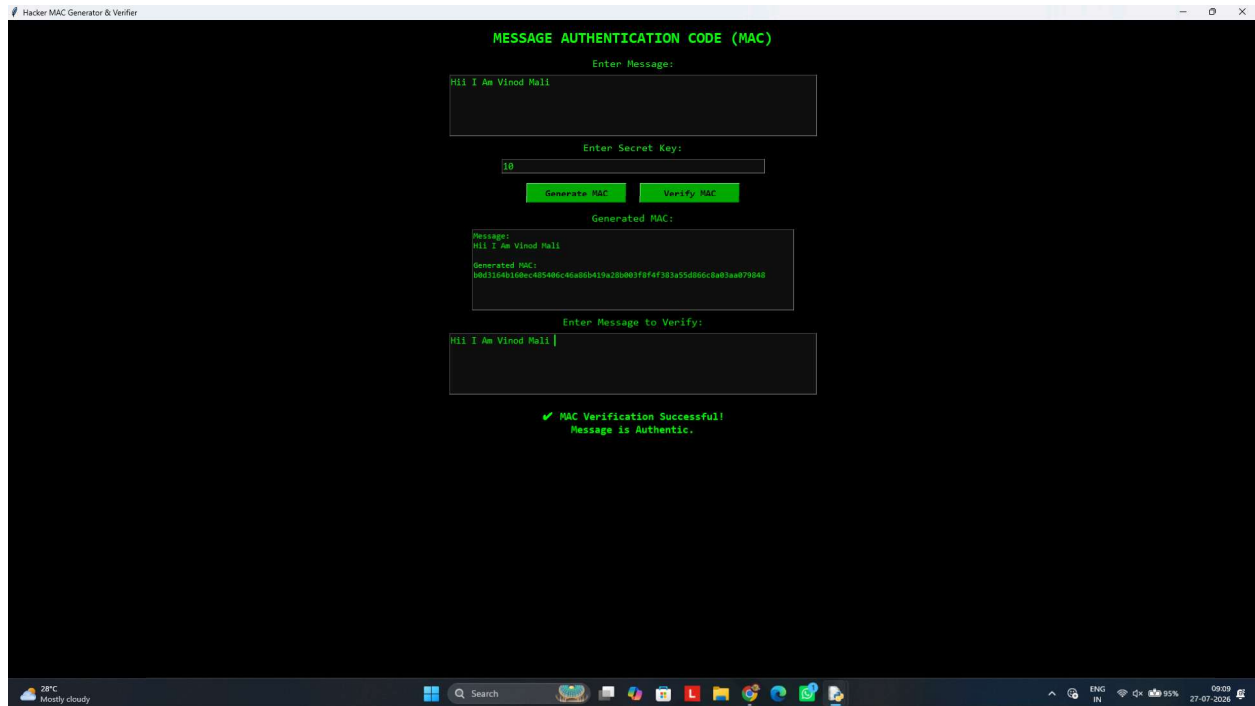
verify_entry = tk.Text(root, height=5, width=70,
                    bg="#101010", fg="#00FF00",
                    insertbackground="#00FF00",
                    font=("Consolas", 11))
verify_entry.pack(pady=5)

# Result
```

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OutPut :-



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